Climate Smart Agriculture techniques as a strategy for Climate Change Adaptation by Smallholder Farmers in Nkayi District.

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Abstract

The study investigates the viability of Climate Smart Agriculture (CSA) techniques being utilized in communal areas in Zimbabwe as mitigation strategy to the effects of climate change, with particular reference to Wards 18 and 19 in Nkayi District. As their livelihoods are highly dependent on climate-sensitive activities such as rain-fed agriculture, climate change has a significantly negative effect on people's livelihoods in communal areas of this country. Increased temperatures, reduced annual rainfall marked by dry spells in the mid-season, and increased frequency and severity of climate-related hazards such as droughts and El Ninos have been detrimental to the livelihoods of communal farmers. The most important problems facing communal farmers in the Nkayi District as a result of climate change are food shortage, water scarcity and livestock loss. CSA techniques are the solution to this predicament. Conservation Agriculture (CA), small grains production, Water Harvesting Techniques (WHTs), fodder production, small livestock production and seed banks are some of the techniques available to communal farmers. Community-based adaptation to climate change helps to develop communities' capacity to guide climate change adaptation strategies to fix their own challenges and to meet their own needs. Thus, traditional awareness, unique environments, tools and mechanisms for societies to map their own strategies to tackle climate change are taken into account by community-based adaptation. The research thus concludes that CSA techniques are a viable strategy in mitigating the effects of climate change.

Key words: climate change, Climate Smart Agriculture, Viability

1. Introduction

The impact of climate change is expected to worsen the already existing challenges faced by developing nations if not dealt with accordingly. Its negative impacts are highly felt by the developing nations who contribute a very insignificant towards its effects. Dube et.al (2016) posit that the irony of climate change phenomenon is that it is the poor countries that have contributed little to the problem of global warming are likely to suffer the most from climate change, since they have the least capacity to adapt to its effects. Climate change impacts have left many communities worldwide vulnerable (Benveniste, Oppenheimer and Fleurbaey 2020). The United Nations Framework Convention for Climate Change (UNFCCC) has highlighted that developing countries are the most vulnerable to climate change impacts because they have fewer resources (socially, technologically and financially) to adapt to climate change impacts (UN 2007).

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Garcia (2008) in Dube et.al (2016) points out that there are three key points that make Africa the most vulnerable continent to climate change. Firstly, the geographical positioning of the African continent gives it one of the warmest climates by virtue of its proximity to the equator. Secondly most African countries depend on the agricultural sector which is sensitive to climate change. Thirdly, the socio-economic gaps in governance, government financing, high rates of poverty and growing populations all expose Africa as a region with vulnerability to climate change. Climate change has far-reaching and unpredictable environmental, social and economic negative impacts. Climate change's widening effects are and will most likely be felt on the environmental and socio-economic sectors such as water resources, agriculture, food security, human health, ecosystems and biodiversity (Ayanlade et.al 2020).

Tropical cyclones, floods, droughts and heavy precipitation are expected to rise even with relatively small average temperatures as a result of global warming and intensity of extreme events Teshome and Zhang (2019). The rise in temperatures will result in the shifts in crop growing seasons which in turn affect food security and changes in the distribution of disease vectors putting more people at risk from diseases (UNFCCC, 2009). This affects the livelihoods of the people as most people will be forced to adapt to these acute vulnerabilities and shocks. Severe floods and drought expose many people particularly for the developing countries to poverty, hunger, disease and in some extreme cases to death. Narrowing down from Africa, Zimbabwe is not spared from the devastating impacts of climate change which have not only increased rainfall variability but have also prolonged dry spells, heat waves, droughts and flooding (Sibanda and Sibanda 2019). In Zimbabwe, climate change effects are felt very much on various sectors inclusive of water, energy, built infrastructure, transport and economic development.

2. Review of relevant literature

2.1 Climate Impacts in Zimbabwe

In Zimbabwe the climate change impacts are mostly related to food security and water supply. Zimbabwe is already prone to droughts which have become recurrent over the past two decades and the geographical location of Zimbabwe in the tropics has makes the country vulnerable to shifting rainfall patterns and water resource availability (Frischen et.al 2020). Therefore, as a result of dependence of Zimbabwe on the rain-fed agriculture, the absence of natural water bodies such as lakes, growing population and the frequent re-occurrence of droughts has led to devastating climate change socio-economic impacts. The consequences of failure to deal with climate change include but not limited to rising sea levels, more frequent heatwaves, coastal flooding, uneven rainfall distribution, more destructive weather events such as cyclones, desertification, vanishing ecosystems, growing and costly health impacts, mass migration, increased conflict and reduced agricultural output (Mashizha 2019)

2.1.1 Climate Impact on Crop Production

Zimbabwe is an agro-based economy. The agricultural sector is a rain fed sector. Due to climate change, recurrent droughts, erratic rainfall and extreme weather events including cyclones (cyclone Dineo and Cyclone Idai) have rendered Zimbabwe's agricultural sector highly vulnerable to climate unpredictability (Mavhura, 2020). FAO (2018) states that Zimbabwe's agriculture heavily relies on 80 percent rainfed crops such as maize, millet, sorghum and wheat. The agricultural sector supports the country's economy, food security and poverty reduction efforts as

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it is the main livelihood for 90 percent of rural households and employs nearly 70 percent of the total population (FAO 2016). Dube and Phiri (2013) posit that rural livelihoods in Zimbabwe, as in most other Sub-Saharan countries mostly revolve around farming activities which are dependent on rainfall. Maize dominates the Zimbabwean agricultural sector and it is cultivated across the country and FAO (2013b) posits that maize is particularly vulnerable to rising temperatures and lack of precipitation as temperatures over 35 degrees Celsius result in lower yields. Therefore, climate change impacts on agriculture results in reduced soil moisture, decreased yields on rainfed crops and crop failure leading to increased food shortages and increase in prices and imports. Impacts on agriculture also leads to increased incidence of pests such as locusts, fall armyworm as a result of increased frequency of droughts (Mafongoya et.al 2019). More so the increase in the intensity of extreme weather events such as cyclones may lead to water logging, flooding which damages crops and leads to crop failure.

2.1.2 Climate impacts on Livestock Production

In Zimbabwe, livestock is an important asset the rural communities possess. Livestock production is a crucial sector in Zimbabwe as livestock is a key source of food, capital, draught power and income, hence the ownership of livestock for smallholder farmers in rural semiarid areas is very important. FAO (2018) postulate that extreme heat, heavy rainfall events (both seasonal and cyclone induced), recurrent droughts and flooding are the most significant stressors facing Zimbabwe's livestock sector which is dominated by cattle, goats, sheep and pigs. Climate change impacts are felt by both small and big livestock. Mambodiyani (2015) states that are the most abundant livestock animal in Zimbabwe with 60-75 percent of households owning cattle. UNDP (2017) reports that during the 2015-16 drought, 27 percent of the reported cattle deaths were drought related due to poor grazing and lack of water. Rurinda (2014) says decreased precipitation combined with heat stress and intense rainfall events also have negative impacts for livestock health, creating conducive conditions that favor the increased incidence and range of livestock pests and diseases such as Newcastle, rift valley fever, theilerioses, zoonoses, anthrax and foot-and-mouth disease. Some of these disease lead to the death of livestock of which communities rely on these livestock for their livelihoods.

2.1.3 Climate Impact on Water Resources

Davis and Hijri (2014) state that country especially in urban areas relies mostly on surface water that is rivers, lakes and reservoirs for its needs of about 90 percent and has almost 40 mediums to large reservoirs and lakes. Therefore, the country coupled with population growth, climate stressors such as rising temperatures leading to higher evaporation rates, reduced precipitation, more intense flooding and more frequent cyclones have diverse and mostly negative impacts on the country's water availability and quality. Reduced river and stream flows as a result of climate change also impact the electricity production within the country. In 2014, water bodies contributed 51.4 percent of national power production (IEA, 2014a) hence reoccurrence of episodes of droughts coupled with variable rainfall patterns have led to reduced water levels and reduced electricity generation from water bodies. Therefore, the rising temperatures lead to reduced water availability resulting reduced crop yields and longer and more dry spells lead to reduced runoff and stream flows impacting people, crops, livestock and hydropower. World Bank (2019) state that the overall reduction in water availability means that by 2080, should medium or high population growth projections be realized, Zimbabwe could fall into the United Nations "absolute water scarcity" category.

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2.1.4 Climate Impacts on Health

It is well understood that human health is adversely by the impacts of climate change (WHO 2017). The World Health Organization estimates an increase of 250,000 excess deaths per year between 2030 and 2050 due to the well understood impacts of climate change (Watts et.al 2015). The increased frequency of floods and heavy rainfall events put already vulnerable communities that lack access to clean water or at greater risk of gastrointestinal diseases such as cholera, typhoid and diarrhea. Mudzonga and Chigwada 2009 say that projected increased temperatures, frequency and or intensity of floods and droughts will adversely impact human health and nutrition across Zimbabwe. Malar et.al (2017) posit that climate stressors will expand the geographic range of malaria and dengue and increase the burden of waterborne and diarrheal diseases like cholera as malaria is the third leading cause of morbidity nationally and by 2050 changes in temperature and precipitation are expected to alter the transmission and increase the incidence of malaria with highest risks of malaria concentrated in the southeastern lower-lying regions and Zambezi river valley. Increased temperatures, unpredictable weather patterns, lead to droughts which will pose a significant threat to food security leading to malnutrition. Malnutrition hence leaves communities vulnerable and susceptible to diseases given the country's heavily reliance on subsistence and rainfed agriculture (Mashizha 2019). Disease have been linked to decreases in household production of staples such as maize, reducing income and the purchasing power. Climate stressors such as rising temperatures lead to increased mortality and morbidity related to heat stress and increased variability of rainfall patterns leads to the expansion of disease carrying vectors such as mosquitoes and increased transmission of infectious diseases such as malaria (Lord et.al 2018).

2.2 Knowledge levels of climate change Rural Zimbabwe

Adequate acquaintance of climate change scenario is indispensable to the success of its global remediation efforts (Ogunbode et.al 2019). Information sharing for evidence-based adaptation to climate change is essential to vulnerability reduction. Even though changes have been occurring over generations, rural farmers have been acclimatizing to these fluctuations throughout their life using home-grown environmental familiarity (Gyampoh et.al 2009; Mafongoya and Ajayi 2017). Environmental problems differ spatio-temporarily, but rural farmers have developed a vast local knowledge of nature in their area through continuous exploration, trial and error, and sustained encounters with their local environment, which they use to cope with and solve their problems, including climate-related problems (Boansi, Tambo and Muller 2017). In a research conducted in Murewa and Seke, results indicated that people in the rural Zimbabwe were aware of climate change, though Murewa had a greater percentage of respondents (94 per cent) than Seke (79 per cent) (Mudombi, Nhamo and Muchie 2014). Local residents have generational awareness of climate changes and natural events and have developed strong adaptation and resilience strategies (Bolden et.al 2018)

2.3 Adaptation strategies to Climate change: challenges and Prospects

Accordingly, adaptation steps to address the impact of climate change are needed to reduce the impact on key sectors, especially agriculture (Iglesias et.al 2012). Climate change adaptation is about making the precise decisions by the communities that will help to mitigate and or reduce the negative effects of climate change. Climate change adaptation as described by UNFCC (2007) is "the process through which the societies increase their ability to cope with an uncertain future, which involves taking appropriate action and making adjustments and changes to reduce negative

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impacts of climate change". Climate change is a terror which has caused havoc within the communities and has led to food insecurity within the communities, hence there is a need for communities to come up with adaptation strategies. Dube et.al (2016) says the adaptation process requires a high level of both technology and resources that communities in the Global South are unable to afford. Adaptation therefore is purposefully needed to reduce the impacts of climate change that is manifesting attached with serious challenges and risks for cities and rural livelihoods such as high temperatures, water availability, floods, droughts and sea-level rise damage to costal zones, Adaptation is therefore vital to climate change impacts on environmental challenges such as soils, biodiversity inland water and marine environment. Adopting effective measures and early actions for climate change adaptation can save and lives. Developing countries have very different individual circumstances and the specific impacts of climate change on a country depend on the climate it experiences as well as its geographical, social, cultural, economic and political situations. As a result, countries require a diversity of adaptation measures very much depending on individual circumstances (UNFCCC, 2005).

Musarurwa and Lunga (2012) postulate that climate change mitigation strategies include policies and activities that reduce the probability of disaster occurrence or reduce the effects of unavoidable disasters. Zimbabwe has established the National Adaptation Plan (NAP) in 2010 with an objective to reduce vulnerability to the impacts of climate change, by building adaptive capacity and resilience and to facilitate the integration of climate change adaptation, in a comprehensible manner, into relevant new and existing policies, programs and activities in particular development planning processes and strategies, within all relevant, economic And environmental sectors and at different levels. The Government of Zimbabwe through the Ministry of Environment, Water and Climate (MEWC) embarked on the NAP process which seeks to develop medium to long term approaches for reducing susceptibility to climate change impacts and facilitate the amalgamation of climate adaptation into ongoing planning processes at national and sub-national planning processes (Government of Zimbabwe 2020) There are thus many climate change adaptation strategies that can be employed. These adaptation strategies can include early warning systems for extreme events, better water management, and improved risk management, various insurance options and biodiversity conservation.

2.3.1 Small grain production

Zimbabwe's farmers create maize, millet, sorghum and wheat to ensure food security. Regardless, maize takes up around 80% – 90% of creation (Zimbabwe Human Development Report 2017). This conveys the country feeble against climate assortment due to the maize reap's sensitivities to ecological change. Small grain crops comprise of plants that produce little usable seeds. Muzerengi and Tirivangasi, (2019), opine that small grains have been noted by research specialists to have a superior yield in dry spell inclined territories and are considered to have preferred dietary substance over maize, which is seen as an unreliable harvest in these farming biological zones. Dry spell safe harvests such sorghum, pearl millet, cowpeas and groundnuts have gotten critical to the nearby network (Ndlovu 2011). This is a result of the way that they go about as both food and money crops, which empowers smallholder ranchers to adjust to environmental change and inconstancy and achieve maintainable jobs (Muzerengi and Tirivangasi, 2019). As a result of small grain production in Mozambique, reasonable yields are achieved even under insufficient rainfall conditions and reasonable yields are achieved even under increasing heat stress (heat) shocks during the growing season that are being experience (Thierfelder et.al 2016). With much

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advantages associated with small grain production, Phiri et.al (2019) refuters that there is a plethora of reasons for the low uptake of small grains as an adaptation strategy to hedge against climate change. One of the main causes of this low uptake is that small grains production in Zimbabwe is affected by high labor costs for land preparation, weeding, bird scaring, harvesting and grain processing compared to maize.

2.3.2 Small Livestock Production

While large livestock mainly refers to cattle, smaller livestock usually refers to sheep, goats, rabbits, ducks, chickens and the like. Recent estimates indicate that Nigeria's national herd comprises 18.4 million cattle, 43.4 million sheep, 76 million goats and 180 million poultry (FMARD, 2017). Phiri et.al (2020) posits that small livestock have higher chances of surviving under climate change conditions like increased droughts compared to bigger livestock since they require smaller grazing land and less amount of water and are less affected by the depletion of pastures. Small stock like goats, sheep, and indigenous poultry are turning out to be predominant as individuals attempt to adapt to dry spell in territories ceaselessly accepting pretty much nothing precipitation (Gukurume 2013). Nonetheless, climate change immensely adds to outrageous temperatures, extreme water shortage, and decreased grain for domesticated animals, particularly huge stock prompting helpless creature creation (Munhande et al. 2013). Climate change also plays a big role in the elevation the risk outbreak and spread of zoonotic diseases, particularly emerging infectious diseases (FAO 2019).

2.3.3 Water harvesting techniques (WHTs)

Water is the primary limiting factor to rain-fed cropping systems under semi-arid ecological conditions, and because of the characteristics of rainfall in general, and low precipitation in particular, the productivity of plants is very low. Ndlovu et.al (2020) notes that in-field rainwater harvesting is one of the adaptation strategies that has been adopted by some smallholder farmers in drought-prone regions. Lasage, and Verburg (2015) adds that water harvesting is widely practiced and is expected to improve water availability for domestic and agricultural use in semi-arid regions. Moreover, Grum et.al (2016) is of the view that water harvesting techniques (WHTs) improve the availability of water, which is essential for growing crops, especially in arid and semi-arid areas. Adjusting climate change by advancing water conservation and administration hones for agribusiness is critical in decreasing trim generation helplessness to climate changeability (Gezie, 2019). Agriculture is practiced mainly under rain-fed conditions, making crop production more vulnerable to climate change and variability in Africa. Climate change has negatively affected areas suitable for agriculture, thus adversely affecting food security and exacerbating malnutrition in the country (Ndlovu et.al 2020). In many regions of Africa, several in-field water management techniques have been used, including earth bunds, pits or planting basins, mulching, dead-level contours and their modifications (Tolossa et al., 2020). However, there are a range of important variables affecting and influencing smallholder farmers' adoption of these water harvesting techniques. These include household labor supply, technological know-how and the expectations of farmers (Ndlovu et.al 2020).

2.3.4 Conservation Agriculture

Kassam, Friedrich and Derpsch (2019) define Conservation Agriculture (CA) as comprising the practical application of three interlinked principles, namely: no or minimum mechanical soil

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disturbance, biomass mulch soil cover and crop species diversification, in conjunction with other complementary good agricultural practices of integrated crop and production management. It increases biodiversity and natural biological processes above and below the ground surface, leading to increased productivity in the use of water and nutrients and enhanced and sustained production of crops. The agriculture sector appears to be at the losing end because of volatile farming conditions, such as short rainy seasons, increasing temperatures, and declining farm fertility, as recorded worldwide, as the world faces daunting climate problems. For more than seven decades, traditional agricultural methods have been championed but remain poorly accepted (Hayat et.al 2019). Above and below the surface of the earth, CA increases biodiversity and natural biological processes. Soil interventions such as mechanical tillage is minimized or avoided to an absolute minimum, and external inputs such as mineral or organic agrochemicals and plant nutrients are implemented optimally and in ways and amounts that do not conflict with or interfere with biological processes (FAO 2014a). Moreover, CA thus encourages good agronomy, such as timely operations, and increases overall land husbandry for the development of rain-fed and irrigated crops. Complemented by other established good practices, including the use of quality seeds and the integrated management of pests, nutrients, weeds and water, etc., CA is the foundation for the intensification of sustainable agricultural production (Kassam et.al 2013; Kassam et.al 2009; Jat, Sahrawat and Kassam 2013; Farooq and Siddique (2014); Friedrich, 2013). Some advantages include, protection of the soil against water and wind erosion; higher productivity and retention of nutrients; less problems and costs with crop protection; and improved efficiency and retention of water usage and improved water economy, including in drylands (Kassam, Friedrich and Derpsch 2019)

2.4 Understanding Resilience

Mitchell (2013) defines resilience as 'the ability of households, communities and nations to absorb and recover from shocks, whilst positively adapting and transforming their uncertainty. The concept of resilience has emerged as a plausible framework among humanitarian and development actors and governments as a longer term and more cost-effective strategy for substantially improving regional improving regional or local capacity to withstand shocks and stresses, ultimately leading to a reduced need for humanitarian response (Mashizha 2019). UNDP (2015) states that building the resilience of vulnerable populations so they can respond positively to potential shocks requires helping people cope with current change, adapt their livelihoods, and improve governance systems and ecosystem health so they are better able to avoid problems in the future. This thus requires the employment of an integrated approach and long term commitment to improving three critical capacities that is absorptive capacity, adaptive capacity, and transformative capacity which are interconnected (Bahadur, Lovell and Pichon 2016), mutually reinforcing, and exist at multiple levels that is individual, household, community, nationally and ecosystem levels (Bene, Newsham and Davies 2012). A mix of all three capacities if often needed to deliver resilient development outcomes in the communities and households. Interventions to increase the absorptive and adaptive capacities are often the first and quickest approach towards increasing the climate resilient communities and individuals.

3. Research Methodology

The research used a case study to gather the information. The case study is an empirical study that explores a contemporary phenomenon in its real-life context and can include both qualitative and quantitative methods of analysis (Yin 1984; Invernizzi et.al 2020). Case studies attempt to

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illuminate a comprehensive understanding of the group, topic and or phenomenon under study. Crowe et.al (2011) opines that the case study approach allows in-depth, multi-faceted explorations of complex issues in their real-life settings. The research uses a case study research design so as to explore the perceptions of climate change within the community of Nkayi district, explore the effects of climate change and investigate the viability of crop diversification in mitigating the effects of climate change. The study was organized into two phases; the field data collection exercise and the key informant interviewing. The field study included the moderation of the focus group discussion. The second phase was conducting three interviews with representatives from the government agency/AGRITEX and the NGO advocating for climate smart agriculture techniques in Nkayi District. The research used a mixed method approach research. Mixed methods approach is the form of research in which a researcher or team of researchers blends elements of qualitative and quantitative research approaches for broad purposes of scope and depth of understanding and corroboration (e.g., use of qualitative and quantitative perspectives, data collection, review, inference techniques) (Schoonenboom and Johnson 2017). The methodology of the mixed method is seen to step beyond the bias between quantitative and qualitative research to a paradigm that appreciates the strengths of both qualitative and quantitative approaches to research. The researcher used both the qualitative and the quantitative analysis method in this approach simultaneously.

4. Research Findings

4.1 Perceptions on Climate Change in Nkayi

Community perceptions and experiences with regards to climate change are a key factor in determining effectiveness of climate change strategies. Community based adaptation draws its strength in building on community experience, and that communities have been managing (positively or negatively) climate hazards for centuries. Contrary to common belief, communities in rural Zimbabwe are aware of their changing environment, and that these communities are not bystanders but active agents for climate change adaptation. Hence, 100% of the communal farmers noted that they have observed changes in weather patterns. The most salient change that was noted by the majority of respondents was the shortening of the rainy season. Increase in temperatures which depleted the water sources and grazing land and frequency of droughts which led to the death of livestock was the next most common response. One responded from pointed that:

> *" Climate change is the change in weather patterns in that rainfall is no longer received in its expected time as in the latter". (Responded 3: Household Interview)*

Most respondents from the household interview and FGD attributed these changes in climate to a range of possible causes, the most common of which were gas emissions, deforestation and the abandonment of traditional/cultural activities, such as picking up of bones in the forests. They expressed that they were aware of climate change and that they have experienced and are still experiencing climate change in the district. The respondents indicated that some of the climate changes noted was the change in rainfall patterns within the district noting that they understand that they are in Region 4 however the optimal rainfall they have received is even below the expectation considering that they are in Region 4. Respondents attributed that they are part and parcel of the causes of climate change because deforestation is a man-made causal. Another responded from echoed that:

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"Partially we as a community are to blame for the emergency of climate change in our community. We cut down trees randomly without care of the effects of doing that. Today we cry but this is partially our own making". (Focus Group Discussion Ward 18)

Another farmer in echoed the same sentiments that:

"We know climate change, and we know that we are partially to blame for its emergency in our community because we cut down trees at the same time not replanting them. We have also abandoned our own culture. Back in the day we used to get into the bush and pick up bones of unknown dead people. Nowadays, we no longer do that. It is a doing of our own". (Focus Group Discussion Ward 19)

Perceptions are affected by norms, levels of education, religion, access to information and historical experiences amongst others which are unique to a particular locality. Key informant responded revealed that participants are better educated with regards to climate change and farmers are aware of climate change. The responded noted that:

"We have done our very best through farmer trainings to empower farmers about climate change and most of them at this level are aware of what climate change is and its effects hence most of them are doing their best to mitigate its effects". (Key Informant A; MELANA)

The other key informant also echoed on the same sentiments that:

"Our role is to empower farmers, and we have and are still continuing in making awareness campaigns with regards to climate change. Most of them are aware of climate change and the effects of climate change". (Key Informant B; AGRITEX)

Participants expressed that they have a better understanding of the climate change. They expressed that as a result of their understanding there is a need to change their traditional way of farming in order to manage climate variability.

4.2 Effects and Experiences of Climate Change

Participants expressed fully that climate change has been a terror with the Nkayi community and has had adverse effects on the livelihoods and largely on the agricultural output since their livelihoods are mainly agro-based. Farmers expressed that effects of climate change include and are not limited to drought, death of livestock, depletion of grazing land for livestock, depletion of water sources leading to scarcity of water, wilting of crops leading to crop failure and shortage of wild fruits generally.

As a result of these effects, respondents that this has forced the community to come up with strategies to try and deal with these effects. One of the major strategies that the respondents expressed was the change in farming seasons accordingly to rainfall patterns of that particular year. A responded expressed that:

"We no longer farm on our conventional farming days but we first look at the rainfall patterns of that particular season, that is why you will note that at times we grow our crops early at times we do late planting" (Focus Group Discussion Ward 19).

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Similar sentiments were raised in Ward 18:

"The farming season has changed in that we wait for the rainfall. We used to start our farming activity late October to early November but now in other seasons we start in early January". (Focus Group Discussion Ward 18)

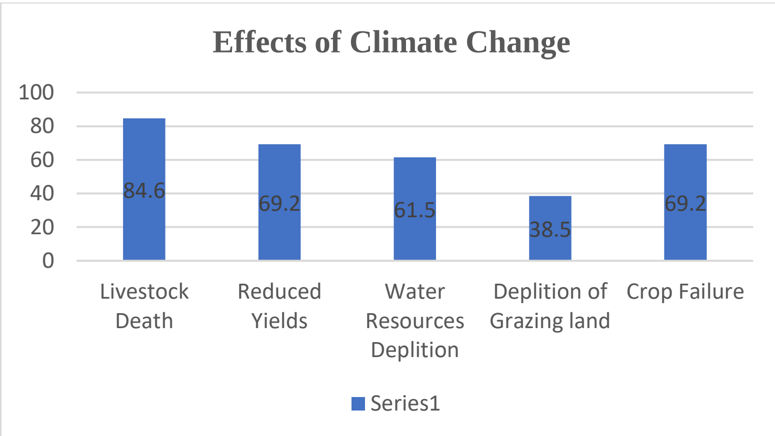

Figure 1: Graph indicating the effects of Climate Change in the communities of Wards 18 and 19

The most common challenges/effects that climate change has posed in Wards 18 and 19 as indicated in Figure 1 above were that of, livestock deaths (84.6%), depletion of water sources (61.5%), reduced yields (69.2%), crop failure (69.2%) and depletion of grazing land (38.5%). Most communal farmers bemoaned decreasing yields over the years.

"I was born in 1938, and yes, we have experienced death of livestock during bad seasons however comparing with the current times, climate change has taken a lot of our pride (livestock) with it. Yields have and are drastically reducing meaning we are doomed towards poverty as households". (Respondent 13; Household Interview)

Another farmer also indicated that:

"We have never experienced such crop failure concurrently in the years. The trend has always been that we have a bad year followed by a good year however it is not the same. Climate change is deadly, it is the mother of poverty. Poverty in that dams are drying up; yields are poor and grazing land is destroyed". (Respondent 5; Household Interview)

Effects of climate change are not only felt in the field but however, there are disruptions in the households. Livelihoods are disturbed. The key informant expressed that there are more challenges posed by climate change than benefits in the communities that the organization is working on. The key informant expressed that challenges included reduced yields, livestock diseases, malnutrition in children, gender- based violence due to droughts, high prevalence of diseases due to droughts and at some extent it has led to spousal separation.

"Climate change effects are felt even beyond the field. Yes, we may talk about yields, droughts death of livestock etc. However, one has to note that poverty as a result of climate change has led to the exacerbation of gender-based violence within the households,

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> *malnutrition in children and mostly most households have been left female and child-headed because of spousal separation". (Key Informant A; MELANA)*

The challenges in the communities in terms of climate change requires the attitudes change of the farmer, transformation of skills, farmers should get new skills and knowledge and transformation in technologies that farmers are using. However, the challenge is that farmers are incapacitated in meeting these requirements since most farmers are dependent on government inputs and or NGO relief handouts, hence there is an imbalance between the government/NGO policy and farmer requirements. The informant expressed that:

> *"Farmers suffer a lot from dependency syndrome. They do not have means to get the inputs for their agricultural season hence they are forced to wait for handouts from either the government (Presidential Inputs Scheme) and or from NGOs of which in most cases these agencies deliver late as they are many procedures for procurement of the inputs. If only farmers could sully embrace seed banking because it is a more viable way to preserve seed". (Key Informant B; AGRITEX)*

Effects and experiences of climate change are thus broad in the communities of Nkayi District. They range from the livelihood base which is agriculture and going further to disrupt households.

4.3 Viability of Climate Smart Agriculture

4.3.1 Knowledge levels and utilized Climate Smart Agriculture techniques

When the researcher sought to find out if the respondents knew about Climate Smart Agriculture, 100% response from the farmers indicated that all the farmers were knowledgeable about Climate Smart Agriculture.

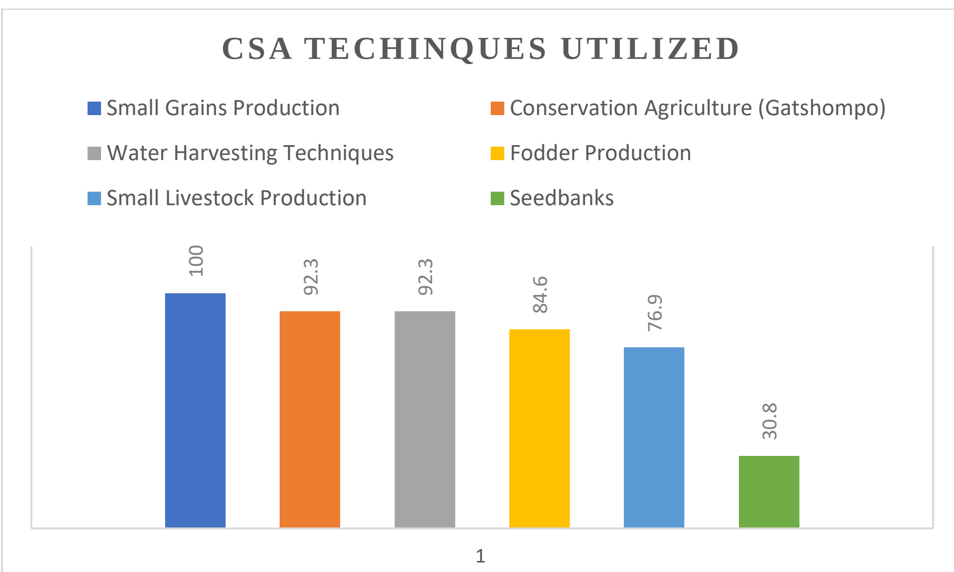

Figure 2: Graph shows the Climate Techniques Utilized

The researcher found that there are a number of climate smart agriculture techniques utilized by Wards 18 and 19 households. These climate smart agriculture techniques are: small grains

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production (100%), conservation agriculture (92.3%), water harvesting techniques (92.3), fodder production (84.6%), small livestock production (76.9%) and seed banks (30.8%). This means that almost all the farmers have found small grains production a viable solution. One farmer mentioned that:

> *"Ever since I was taught about climate smart agriculture, I have made it an obligation that I grow small grains without fail. The small grain varieties that I normally grow are millet and sorghum and frankly they have not disappointed me in terms of yields and I can assure you they are drought tolerant". (Respondent 8: Household interview)*

The study revealed that there was evidence that the community was being empowered to take a leading role in implementing Climate Smart Agriculture techniques. The participants expressed that they received trainings on climate smart agriculture techniques and that they were responsible for the implementation of what they are trained on. As mentioned by the respondents in the household survey, participants alluded that selection of the climate smart agriculture techniques was chosen by a collaboration of the community together with government extension workers (AGRITEX) and NGO (MELANA).

> *"MELANA and AGRITEX officers have been wonderful teachers to us with regards to climate smart agriculture. They have empowered us to grow crops that we had regarded as useless at some point such as millet and sorghum. We have gone back to what we grew up growing. We were taught to practice conservation farming (gatshompo) and it is helping in improving our yields". (Focus Group Discussion Ward 18)*

The participants expressed that the techniques that they have adopted, include small grain production, small livestock production, conservation agriculture, fodder production, water harvesting techniques and seed banks. As mentioned by respondents in the household questionnaire survey, the participants revealed that conservation farming and small grains production had helped increase their food security at household level, fodder production has helped in boosting their livestock.

The key informant respondent echoed that the organization has put in place climate smart agriculture techniques through the Resilience Building Programme which the organization is implementing. The programs implemented include small grains production, small livestock production, fodder production, conservation agriculture, water harvesting (through basins, dead level contours), seed banks, social safety nets (ISALS) and crop diversification. The responded expressed same sentiments like the household respondents and focus group participants that farmers have been taught about climate smart agriculture. The responded posited that:

> *"Through the Resilience Building Programme, we have empowered farmers about climate smart agriculture in partnership with the government agencies (Agritex and Veterinary services) through the trainings we have conducted in the field. We have even established Farmer Field Schools and Demo-plots where farmers are taught practically". (Key Informant A; MELANA).*

This was echoed by the other key informant who alluded that:

> *"We are currently are implementing the Intwasa/Pfumvidza program in collaboration with the NGO sector. The Intwasa program is focused on minimum disturbance of the land and*

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> *use of conservative agriculture methodology (gatshompo). Crop diversification, small grains production (drought tolerant crops), water harvesting techniques and enterprise diversification (fodder production, small livestock production and agroforestry crops) are other programs that have and are still implemented as climate smart agriculture techniques". (Key Informant B; AGRITEX)*

4.3.1.1 Conservation agriculture

Conservation agriculture aims to achieve sustainable and profitable agriculture, and subsequently aim to improve livelihoods of farmers as a climatic adaptation strategy through the application of the three-conservation agriculture (CA) principles, which are minimal soil disturbance, permanent soil cover and crop rotation. The respondents from both the questionnaire survey and the focus group discussions purported that CA had potential to improve their food security at household level, regardless of the size of field that they had. It is a way to combine profitable agricultural production with environmental concerns and sustainability

4.3.1.2 Small grains production

The respondents from the household questionnaire survey and focus group discussions highlighted that small grains production was a strategy that they had been using for generations. They stated that they had been growing millet, sorghum and rapoko, so as to reduce the impact of drought.

4.3.1.3 Water harvesting techniques

Water harvesting is undertaken to capture the runoff from the seasonal rains and store it in times of need. When the rainfall is scant, part of it gets lost during interception, evaporation and runoff leaving little for storage and future use. Therefore, the respondents from the focus group discussions put forward that rainwater harvesting as a climate adaptation strategy is essential to them for effective use by the household, livestock and agriculture.

4.3.1.4 Small Livestock Production.

The respondents from the household questionnaire and focus group discussion highlighted that small livestock production was a strategy that they have lately adopted. They now focus on the rearing of small livestock such as Chickens, goats and pigs as these livestock are less vulnerable to climate change as compared to large livestock such as cattle.

4.3.1.5 Fodder Production

Respondents expressed that as a mitigation strategy they now focus on fodder production. They expressed that they now grow crops for livestock. Fodder crops commonly grown include velvet bean, Lab-lab and Forage sorghum. One respondent from a household questionnaire survey had this to say:

> *"My child, climate smart agriculture has taught us that our livestock is part and parcel of our family, hence there is need to also grow crops that belong to the livestock as they are an important part of the family. We used to treat our livestock as workers but now we treat our livestock as family hence we are so much engaged in fodder production" (Focus Group Discussion Ward 18)*

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4.3.2 Community participation in climate change adaptation strategies

When the researcher sought to investigate the relationship between participation and effectiveness of climate smart agriculture techniques in communities, the response solicited from respondents expressed mixed views. When asked how the communal farmers were selected for NGO or government-initiated climate smart agriculture programs, the following views were expressed:

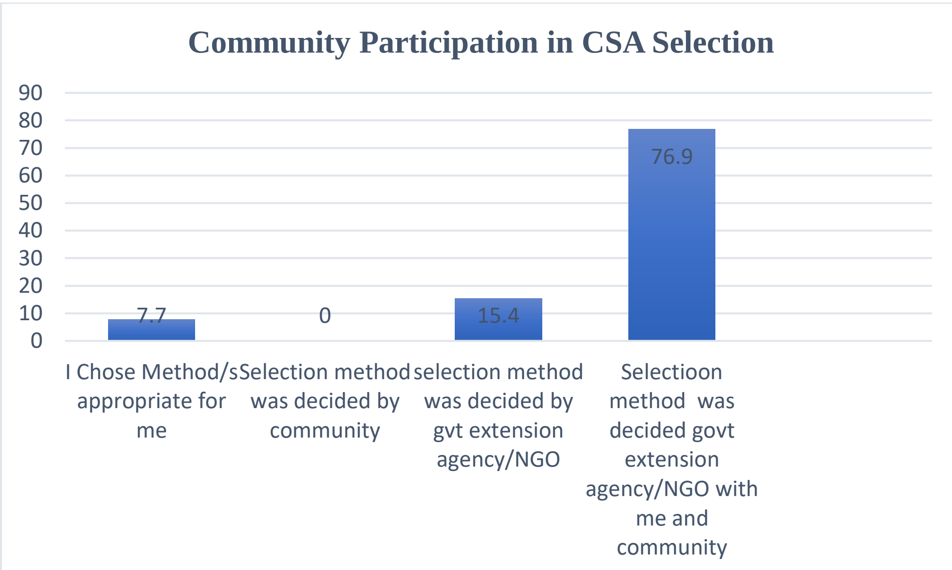

Figure 3: Perception on participation/decision making on selection criteria responses by percentage.

The majority of responses (76.9%) indicate that that the selection criterion was decided by the implementing agencies in the community with the community. Similarly, when asked how the climate smart agriculture techniques were selected, the majority of respondents expressed that the climate smart agriculture techniques that they were utilizing were selected by the government or NGO agency with community.

> *"We were called for a community meeting by Agritex and MELANA, and we were informed about these techniques and together with them we chose to use these method". (Respondent 12; Household interview)*

However other farmers solicitated that the implementing agencies selected the techniques for them alluding that:

> *"These techniques were chosen for us since they are the only techniques, they taught us besides they were not giving us much of a choice we had to comply. I cannot complain however because these techniques are giving us good results". (Respondent 2; Household Interview)*

The key informants as well not that they collaborated with farmers on choosing the techniques appropriate:

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"We engaged the farmers (community) together with MELANA and prior to choosing the CSA techniques implemented. The community was happy and has been implementing what we teach them". (Key Informant B; AGRITEX)

The community-based adaptation approach (CBAA) advocates the facilitation of a learning process that increases resilience and problem-solving capacity. The responses indicate that the communities of Wards 18 and 19 were directly involved in the decisions concerning the climate change smart agriculture techniques they are utilizing.

4.3.3 Effectiveness of CSA techniques

The study revealed that all the respondents felt that the strategies they were employing to combat climate change were helping them to a greater extent. The majority of respondents stated that their household food security had improved significantly. Practicing conservation farming had increased their yields, and cultivation of small grains had improved their cereal sufficiency with the declining year on year harvest from maize.

"I can assure you that yields are better ever since I started using conservation agriculture (gatshompo) and growing small grains. My livestock is also benefiting from fodder production. Poverty is slowly becoming a thing of the past". (Respondent 13; Household Interview)

As mentioned by respondents in the household questionnaire survey, the participants revealed that conservation farming and small grains production had helped increase their food security at household level, fodder production has helped in boosting their livestock:

" We cannot complain much we are reaching food security slowly. Small grains are tolerant to our harsh weather patterns and we are able to harvest something. Our livestock is also benefiting out of this. We are grateful to CSA". (Focus Group Discussion Ward 19)

The key informant alluded that climate smart agriculture has increased yields, reduced movement of farmers to search for food as they can buy locally from each other (reference to stock feed produced locally by the Bushmeal), income increase as farmers buy from each other and helping vulnerable households through community nutrition gardens. The responded expressed that:

"Climate smart agriculture is one of the most successful programmes that they have implemented as an organization as it has positively yielded and is building resilience amongst the farmers in the district. I can attest that even though there are challenges they face but they are outweighed by the success of CSA techniques" (Key Informant A; MELANA)

Yields have improved from various crops especially small grains, reduced risk of crop failure, with water harvesting techniques farmers now store water and enterprise diversification has seen farmers increasing incomes through selling of eggs from poultry production and they use the money to buy other necessities.

4.3.4 Challenges of CSA.

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Despite the advantages of CSA, 53.8% of the respondents revealed that they do face challenges with regards to implementing climate change adaptation strategies, in comparison with 46.2% of respondents stating that they faced no challenges. The study revealed that most challenges that communal communities face with regards to climate smart agriculture techniques were that it was labor intensive especially with regards to conservation farming, quelea birds especially with regards to small grains and emergency of new pests such as the fall-army worm.

> "*Well what can we say… nothing comes easy. Small grains come with birds. We have a problem of birds but well it is how God created it. They have to eat in our fields. Conservation farming is very labor intensive in-terms of digging planting holes. We face these challenges". (Respondent 1; Household interview)*

This was echoed by the participants from the FGD:

> *"Small grains production is associated with problems such as quelea birds and also that it is labor intensive. It is very labor intensive particularly during harvesting, more so we have since realized the emergency of new pests such as fall-army worm which destroys the crops" (FGD Ward 19)*

They also mention the problems associated with quelea birds which were very problematic particularly on small grains. The key informant also added that:

> *"The main challenge realized since the utilization of these CSA techniques is that they may be labor intensive, issues to do with quelea birds and that farmers are laggards (adoption rate is very slow). Hence on that regard the organization continues conducting trainings and awareness campaigns as a supportive measure". (Key Informant A; MELANA)*

4.4 Discussion

The verdicts in this paper depiction new and evolving evidence of the link between practicality of climate smart agriculture techniques and climate change whereas at the same time reverberating findings from other studies. With regards to the perceptions on climate change, studies by Ndlovu et.al (2020) allude that "an understanding of smallholder farmers' perceptions of climate change is important in analyzing the adaptive strategies they adopt". Hence in the case of this study, farmers are well aware of climate change and it effects hence it has influenced their adoption of the climate smart agriculture techniques as adaptive measures towards the effects of climate change that they encounter on their day-to-day lives.

On the effects of climate change farmers express that the effects are diverse ranging from high temperatures, death of livestock, change in rainfall patterns. In a study conducted in Zambia similar sentiments by the farmers are encountered. Mulenga, Wineman and Sitko (2017) posit that "farmers offer remarkably consistent reports of a rainy season that is growing shorter and less predictable". More so as mentioned by the respondents, climate change has posed a great threat to water availability. Climate change has exacerbated the shortage of water. Water bodies are drying up and water as an important resource is becoming scarce. Hamududu and Ngoma (2020) in a study carried out in Zambia posit that, "in particular, Zambezi, Kafue and Luangwa River basins are projected to have less water resources available due to reduced rainfall and higher temperatures."

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One of the major noted benefits of adopting CSA techniques is the improvement in yields as noted by the farmers. In a study conducted in Malawi same sentiments are highlighted by Amadu, McNamara, and Miller (2020) who posit that "we found a 53% increase in maize yield among CSA adopters in the drought year of 2016. Results demonstrate that policies and funding streams supporting CSA in low-income, dryland contexts such as southern Malawi can have important impacts on food security by boosting crop yields in the face of increasing climate uncertainty and extreme weather shocks."

With regards to the challenges of the CSA techniques such as small grains production, they are very labor intensive such that as much as more farmers engage in small grains production, they feel burdened. Similar studies by Dube et.al (2018) reveal that small grains production is a very labor-intensive strategy that is being abandoned in Gwanda. Small grains as well is affected by quelea birds as highlighted by all the respondents. Similar studies by Phiri et.al (2019) echo that "the red-billed quelea bird is a major pest of small grain crops in Zimbabwe both on irrigated wheat and barley in the winter and on sorghum and millet in the summer. The red-billed quelea birds have been a continuing challenge on small grain crops in Zimbabwe".

5. Conclusion

The study concludes that climate smart agriculture as a strategy to climate change adaptation is viable. The multi-faceted impacts of climate change faced by smallholder farmers are horrifying to tolerate. The findings indicate that the smallholder farmers are crippled by drought, death of livestock, depletion of water resources, depletion of grazing land and reduced yields. These negative impacts have heard a bearing on household disturbances such as gender-based violence and to some extent spousal separation. However, smallholder farmers have adapted to the negative undulation effects of climate change through climate smart agriculture techniques that include small grains production, conservation agriculture, small livestock production, water harvesting techniques and seed banking.

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